The LQ Digital Mode Family: Protocol Architecture and Reference Specification for Weak-Signal Communications

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***Abstract*—This paper introduces LQ8, an open-source weak-signal digital mode for amateur radio that completes structured two-way contacts in 4 transmissions (60 s), achieving a 1.50× speedup over canonical 6-transmission exchanges (90 s) and 1.25× over 5-transmission `RR73` exchanges (75 s) while preserving complete bidirectional exchange of Maidenhead grid locators, signal reports, and mutual acknowledgments. By replacing fixed-width message-type headers with variable-length prefix codes matched to physical-layer bit budgets, LQ8 packs the answering station's grid locator and measured signal report into a single 77-bit reply. In multi-station pileup operations, LQ8 confirms up to two answering stations simultaneously within a single-carrier 50.0 Hz transmission with a 0.0 dB RF power-splitting penalty, achieving a peak rate of 240 QSOs/h in continuous pileup (4× over single-carrier FT8; 480 QSOs/h dual-carrier) while retaining −21.0 dB SNR sensitivity. The protocol family also defines three extended profiles (LQ16, LQ4, and LQ2) tailored for extreme sensitivity, VHF/UHF contesting, and fast-burst channels.**

I. INTRODUCTION

The FT8 digital mode [1], designed by Steve Franke (K9AN) and Joe Taylor (K1JT), has transformed amateur radio since its introduction in 2017 in the WSJT-X software suite [2], enabling reliable contacts at signal-to-noise ratios as low as −21.0 dB in a 2500 Hz bandwidth. Its 77-bit message payload, combined with a low-density parity-check (LDPC) forward error correction code [5] and 8-tone Gaussian frequency-shift keying (8-GFSK), provides a robust physical layer that is now well-proven across millions of contacts worldwide.

In conventional weak-signal protocols (e.g., FT8), a structured two-way contact canonically requires six sequential transmissions across 15-second time slots (90 seconds total), or five transmissions (75 seconds) when using shortened `RR73` confirmations (CQ → CALL → REPORT → RREPORT → RR73). Under a 77-bit payload budget, encoding two 28-bit standard callsigns, a 15-bit Maidenhead grid locator, and a 5-bit signal report requires 76 information bits ($28+28+15+5=76$). Uniform fixed-width type headers (typically 3 or 4 bits) exceed the remaining 1-bit margin, preventing the simultaneous transmission of locator and report in a single transmission.

LQ8 resolves this constraint by employing **variable-length prefix codes** [4] tailored to physical-layer bit budgets. By allocating a 1-bit prefix to the dominant response message format, LQ8 preserves 76 bits for payload data, packing both the answering station's grid locator and measured signal report into a single frame. This completes confirmed contacts in 4 transmissions (60 seconds) with full mutual verification, while retaining the proven continuous-phase 8-GFSK and LDPC$(174,91)$ physical layer (−21.0 dB SNR sensitivity). Furthermore, for high-rate DXpedition Fox and Hound (F/H) pileups, LQ8 introduces the MULTI-REPORT+73 protocol to confirm two answering Hound stations simultaneously within a single-carrier transmission without RF power-splitting penalties. Cross-isolated Costas synchronization arrays ($C_7=[2,5,6,1,3,0,4]$, exhibiting ≤ 1 coincidence hit) isolate traffic from legacy decoders. Physical layer, CRC-14, and LDPC parameters are presented in Section V.

Building upon LQ8, the protocol family defines three complementary profiles (LQ16, LQ4, and LQ2) tailored for ultra-weak fading channels, rapid contesting, and fast burst links.

A. Architectural Philosophy & Design Principles

The LQ family is engineered around four core design tenets:

- **Payload-Matched Prefix Allocation:** Replacing uniform headers with variable-length prefix codes frees capacity for simultaneous locator and report transmission, shortening 1-on-1 contacts to 4 slots (60 s) and reducing vulnerability to ionospheric fading (QSB) and transient interference.
- **Coexistence and Spectrum Preservation:** All modes maintain continuous-phase GFSK and LDPC$(174,91)$ channel coding while employing low-cross-coincidence Costas sync arrays to prevent cross-protocol false decodes.
- **Robust Special-Case Support:** Non-standard callsigns (up to 13 characters), dedicated 1-bit portable suffixes (`0`=*none*, `1`=`/P`), 24/20-bit collision-resistant callsign hashes, and a 104-character free-text Varicode provide seamless support for contesting, portable activations (POTA/SOTA), and DXpeditions.
- **Clear-Text Station Identification:** Transmitting stations convey their callsigns in clear text across all 1-on-1 exchanges (including non-standard caller replies via 48-bit Base-38 encoding), reserving compact hashes strictly for target addressing and multi-station pileup frames.

B. Implementations & Reference Software

The complete suite is implemented in the open-source C++ reference library (`lq_lib`) [11] (MIT License), featuring an automated verification harness and golden vector test suites across all four modulation profiles. Web demonstrations and interactive protocol resources are published on the official project portal [12]. Native portable operation is provided by **qFT8** for Android [13], deploying LQ8 alongside live testbeds for LQ16, LQ4, and LQ2.

II. QSO PROTOCOL

The LQ protocol establishes deterministic state transitions, transmission slot timing, and multi-station arbitration to achieve confirmed bilateral contacts across low-SNR channels.

A. Standard Four-Message Exchange

The LQ QSO protocol is built upon two core principles: bidirectional exchange of Maidenhead grid locators and signal reports (measured SNR in dB), coupled with mutual confirmation (Table 1). Following weak-signal convention, LQ establishes an operational termination criterion: each station must hear an explicit confirmation after delivering its own signal report before considering the contact complete. Station A considers the contact complete upon receiving Station B's concluding 73 in T4, while Station B considers it complete upon decoding Station A's report in T3 and transmitting its final 73.

Table 1: QSO sequence comparison between conventional protocols (FT8) and the LQ8 4-slot protocol.

Seq	Conventional Protocols (FT8)	Seq	LQ8 Protocol
1	`CQ YO1YO JN47`	1	`CQ YO1YO JN47`
2	`YO1YO TU2TU KL22`	2	`YO1YO TU2TU KL22 -03`
3	`TU2TU YO1YO +05`	3	`TU2TU YO1YO R+05`
4	`YO1YO TU2TU R-03`	4	`YO1YO TU2TU 73` *[Confirmed]*
5	`TU2TU YO1YO RR73` (or `RRR`)	—	—
6	`YO1YO TU2TU 73` (optional if `RR73`)	—	—

As summarized in Table 1, LQ8 completes a contact in 4 time slots (60 s vs. 5–6 messages / 75–90 s in conventional exchanges ending in `RR73` or `RRR`/73), delivering full bidirectional callsigns, Maidenhead locators, and signal reports with mutual verification:

1. **T1 (Slot 1, Even):** Station A transmits a CQ broadcast (catalogued in Section III, Types 1–4) containing its callsign, optional portable suffix, grid locator, or CQ modifier.
2. **T2 (Slot 2, Odd):** Station B transmits a directed CALL reply (Types 5–7) packing Station A's target address, caller callsign, measured SNR report, and grid locator into a single frame.
3. **T3 (Slot 3, Even):** Station A acknowledges with REPORT+73 (Type 8 for standard callsigns, or multi-station Type 11), delivering its measured SNR report for Station B.
4. **T4 (Slot 4, Odd):** Station B transmits a final 73 acknowledgment (Types 9 or 10; Type 12 for 10–13-character compound callsigns), completing the confirmed bidirectional contact.

B. Retransmission, Error Recovery and Logging Semantics

To ensure session progress over lossy channels, LQ formalizes timeout recovery and logging rules.

Retransmission and Repeat Calls. If Station B misses REPORT+73 in T3 (due to QSB or interference), it repeats CALL in the next odd slot. Receiving this repeat informs Station A that its report was lost, prompting Station A to resend REPORT+73. Similarly, if Station A misses the final 73 in T4, it retransmits REPORT+73; Station B reacknowledges with 73 without duplicate logging.

QSO Logging Semantics. Contacts are logged once callsigns, locators, reports, and acknowledgments are exchanged: Station A logs upon decoding the final 73 in T4 (or transmitting REPORT+73 in Fox pileup mode); Station B logs upon transmitting 73 in T4 (having decoded Station A's report in T3). In ADIF logging (LoTW, QRZ.com, Club Log), contacts map to mode `DATA` with submode `LQ8` (`LQ4`, `LQ2`, `LQ16`). A watchdog aborts after three unacknowledged attempts (45 s in LQ8, 22.5 s in LQ4), returning to idle monitoring.

C. DXpedition Fox and Hound Multi-Station Pileup Operation

During DXpedition pileups or contest operations in Fox and Hound (F/H) mode, Station A (Fox) activates the MULTI-REPORT+73 protocol (Type 11) to confirm up to two answering Hound stations simultaneously on a single carrier:

1. **Slot 1 (Fox):** Fox broadcasts directed CQ on its primary run frequency (e.g., `CQ DX FOX`).
2. **Slot 2 (Hounds):** Remote Hounds call simultaneously on split audio frequencies via directed CALL (e.g., `<FOX> <HOUND1> FN42 -08`).
3. **Slot 3 (Fox):** Fox confirms both Hounds simultaneously on the run frequency via MULTI-REPORT+73 (`<HOUND1> R+05 <HOUND2> R-02 <FOX>`).
4. **Slot 4 (Hounds):** Both Hounds respond with concluding 73 acknowledgments on their split frequencies (`<FOX> <HOUND1> 73`).

When confirming a single Hound in Fox mode, standard callers are confirmed using Type 8 (REPORT+73 `STD+SUF`). Single-target Type 11 (or Type 12 for final acknowledgments) is used when confirming non-standard callers or when operating under strict split-frequency F/H scheduling; in this case, Target 2 duplicates Target 1 (identical hash, and identical SNR in Type 11), which decoders process as a single-station transmission.

Throughput Multiplier Analysis. In continuous pileups where Step 3 overlaps with incoming calls, completing 2 QSOs every 2 slots (30 s) delivers a peak rate of 240 QSOs/h (4× over single-carrier FT8; 480 QSOs/h dual-carrier; 160 QSOs/h in isolated 3-slot exchanges) with **0.0 dB power penalty**, avoiding the −3.0 dB RF loss of multi-carrier FT8 while halving fading exposure.

D. Multi-Caller Contention and Pileup Queuing

When multiple stations answer the same CQ simultaneously in 1-on-1 mode, Station A selects one caller and transmits REPORT+73 addressed to that station; other callers retry in the next alternating time slot. Frames are disambiguated via uniform 24-bit callsign hashes ($2^{24} \approx 16.78 \times 10^6$ bins, Section IV-D) and canonical precedence (Section III-C).

To optimize pileup resolution, transceivers implement opportunistic candidate queuing (up to 16 stations). Priority aging balances throughput with fairness, preventing high-SNR arrivals from starving marginal callers. Upon decoding 73 in T4, Station A immediately directs REPORT+73 to the highest-priority queued station without an intermediate CQ, accelerating turnover. If the queue is full, lowest-SNR entries below decode threshold are evicted first; entries expire after four unacknowledged cycles (60 s in LQ8, 30 s in LQ4).

III. MESSAGE ARCHITECTURE

To maximize channel efficiency within the 77-bit physical frame budget, LQ replaces fixed-width bit fields with an entropy-matched variable-length message architecture.

A. Design Motivation & Prefix-Free Framing

Under a strict 77-bit physical payload budget, conventional fixed-width message headers impose an unnecessary penalty on high-entropy transmissions. In a standard CALL reply, encoding two 28-bit standard callsigns, a 15-bit Maidenhead grid locator, and a 5-bit measured SNR report requires 76 information bits:

$$\underbrace{1\text{b}}_{\text{Prefix}} + \underbrace{28\text{b}}_{\text{Target}} + \underbrace{28\text{b}}_{\text{Caller}} + \underbrace{15\text{b}}_{\text{Locator}} + \underbrace{5\text{b}}_{\text{SNR}} = \mathbf{77}\ \text{bits}. \quad (1)$$

Uniform 3-bit or 4-bit type headers exceed the remaining 1-bit margin, preventing simultaneous transmission of locator and report in conventional protocols.

LQ resolves this constraint by synthesizing variable-length prefix coding [4, 3] with physical-layer bit budgets. Assigning a **1-bit** codeword (`1`) to the dominant CALL STD (NO-SUF) format preserves 76 bits for payload data, combining station identification, locator delivery, and signal report measurement into a single frame. Simpler messages (such as standard CQ, requiring 43 payload bits) readily accommodate longer prefix codes.

Formally, the prefix code $\mathscr{C} = \{c_1, c_2, \ldots, c_k\}$ is *prefix-free*:

$$\forall i \neq j: \quad c_i \text{ is not a prefix of } c_j. \quad (2)$$

The receiver traverses a binary prefix tree bit-by-bit from the most-significant bit until reaching a terminal leaf node, identifying the message type without length fields or delimiters. Although the mathematical binary prefix tree across all 16 slots satisfies $\sum_{i=1}^{16} 2^{-\text{len}_i} = 1.0$, the active operational codebook is intentionally prefix-free without being Kraft-complete because Types 14–16 are explicitly reserved for future backward-compatible protocol expansions rather than active operational message formats.

B. Operational Categories & Message Catalogue

Stations select transmission formats according to the operational categories defined in Table 2. Table 3 catalogues all 16 message slots with their prefix codewords, payload layouts, and bit budgets. Frame payloads ≤ 77 bits are zero-padded to the 77-bit link-layer boundary. Payloads are serialized in big-endian bit order (MSB first) into a 10-byte buffer: bit 0 corresponds to `0x80` of Byte 0, bit 7 to `0x01` of Byte 0, and bits 77–79 (the three least significant bits of Byte 9) are set to zero.

Field Designations in Table 3:

- **Canonical Short Identifiers:** Standardized mnemonics: **CQ**, **CALL**, **RPT73** (for REPORT+73), **73**, **RPT73M** (for MULTI-REPORT+73), and **73M** (for MULTI-73).
- **Prefix & Standard Callsign:** "Code" denotes prefix length; "call:std" is a 28-bit standard callsign (Section IV-A).
- **Non-Standard Callsigns & Suffixes:** "call:*n*" is an *n*-character non-standard callsign packed via Base-38 (Section IV-C); "Suf" encodes 1-bit portable modifiers (`0`=*none*, `1`=`/P`). Type 5 omits suffixes to pack two standard callsigns, locator, and SNR into 77 bits ($1+28+28+15+5=77$b).
- **Callsign Hashes:** "Hash" is a 24-bit CRC-24/Q hash ($2^{24} \approx 16.78 \times 10^6$ bins); "hash†" is the 16-bit DX hash (Section IV-D).
- **Multi-Station Fields:** Type 11 Fox pileup sequences [Prefix_3 | DX_{16} | Tgt1_{24} | SNR1_5 | Tgt2_{24} | SNR2_5] (77 bits); Type 12 formats [Prefix_8 | DX_{16} | Tgt1_{24} | Tgt2_{24} | 00000_2] (77 bits, 5 zero-padded bits).
- **Modifiers, Locators & SNR:** "Mod" is a 20-bit modifier (Section IV-F); "loc" is a 15-bit locator (Section IV-E); "SNR" is a 5-bit report (Section IV-G).

C. Protocol Boundary Conditions & Canonical Precedence

To eliminate ambiguity across station profiles, LQ formalizes eight canonical precedence and boundary rules:

1. **Strict Canonical Precedence:** Transmitters **must** use the most compact format fitting caller and target: for CALL, Type 5 requires both stations standard without suffix; Type 6 applies if either has `/P` or target is non-standard; Type 7 if caller is non-standard. For replies, Type 8 precedes Type 11 and Type 9 precedes Types 10/12 if both are standard.
2. **Deterministic Hash Disambiguation:** In Type 6, target hash matches against unsuffixed standard stations are rejected only if caller also lacks a suffix (Type 5 was mandatory); suffixed callers and Type 7 validly target standard stations. In 1-on-1 contacts, standard stations ignore hashed completions (Types 10, 11) from standard peers.
3. **Universal Locator and SNR in Type 6:** Type 6 allocates 4 prefix bits and 24 target hash bits, leaving 49 bits for caller callsign, suffix, 15-bit locator, and 5-bit SNR ($4+24+28+1+15+5=77$b). Suffixed callers deliver locator and SNR on their initial call without extra steps.
4. **Clear-Text Identity in Type 7:** Type 7 encodes the target via 20-bit hash ($H_{20} = H_{24} \gg 4$), the non-standard caller via clear-text 48-bit Base-38 representation (≤ 9 characters), 1-bit suffix (`/P`), and 5-bit SNR ($3+20+48+1+5=77$b), delivering unhashed identity directly in Step 2.
5. **Single-Target Multi-Station Framing:** When confirming one station in Fox mode (Types 11/12), Target 2 duplicates Target 1 (identical hash, and SNR in Type 11), decoded as a single station. In 1-on-1 contacts, Type 8 strictly supersedes Type 11; single-target Type 11 is reserved for non-standard callers or split F/H schedules.
6. **Collision Resistance:** Uniform 24-bit callsign hashes ($2^{24} \approx 16.78 \times 10^6$ bins) yield $P_{\text{collision}} \leq 0.75\%$ across 500 stations, disambiguated via local callsign caches and canonical precedence.
7. **Compound Callsigns:** Stations with callsigns exceeding 9 characters (up to 13 characters, e.g., `3B9/HB9IPH/P`, Category D) utilize the 69-bit Base-38 field in CQ frames (Type 4); directed replies use CQ or free text (Type 13).
8. **Verification & Compatibility:** Bit-exact codec traces and golden vectors across all 13 message formats are verified in the reference suite [11] across AWGN, Rayleigh fading, and clock-drift conditions. Compliant decoders treat unassigned opcodes (Types 14–16) as valid framing boundaries, safely ignoring them.

Table 2: Operational categories and corresponding 4-step message sequences.

Cat.	Callsign / Mode Configuration	Step 1: CQ	Step 2: CALL	Step 3: REPORT+73	Step 4: 73
A	Standard (no suffixes)	Type 1	Type 5	Type 8	Type 9
B	Standard + std suffix (`/P`)	Type 1	Type 6	Type 8	Type 9
C	Non-standard (≤ 9 chars) + optional suffix	Type 2 / 3	Type 7* / Type 6	Type 11	Type 10
D	Long compound (10–13 chars) + optional suffix	Type 4	—‡	Type 11	Type 12
E	Multi-station reply	—	—	Type 11 (Fox)	Type 9 / 10 (Hounds)§

*In Step 2, Category C callers transmit Type 7 (clear-text callsign, suffix, SNR to target 20-bit hash); standard callers answering a Category C CQ transmit Type 6 (locator, SNR to target 24-bit hash). ‡Category D compound stations (10–13 chars) initiate contact via CQ (Type 4) or free text (Type 13). §A Hound with a 10–13-character callsign sends Type 12; the Fox may send an optional MULTI-73 (Type 12) after the exchange.

Table 3: Complete LQ message-type catalogue.

Huffman Code	ID	Type	Bits (Payload)							Total
			Code	Target	Suf	Caller	Suf	Modifier	Locator	SNR	
`0000000000`	1	CQ std+suf	10	—	—	28 call:std	1 suf	20 mod	15 loc	—	74
`0000000001`	2	CQ non-std 1	10	—	—	48 call:9	1 suf	—	15 loc	—	74
`00000001`	3	CQ non-std 2	8	—	—	48 call:9	1 suf	20 mod	—	—	77
`0000001`	4	CQ non-std 3	7	—	—	69 call:13	1 suf	—	—	—	77
`1`	5	CALL std (nosuf)	1	28 call:std	—*	28 call:std	—*	—	15 loc	5 SNR	77
`0100`	6	CALL std+suf	4	24 hash	—	28 call:std	1 suf	—	15 loc	5 SNR	77
`001`	7	CALL non-std	3	20 hash	—	48 call:9	1 suf	—	—	5 SNR	77
`0000000010`	8	REPORT+73 std+suf (RPT73)	10	28 call:std	1 suf	28 call:std	1 suf	—	—	5 SNR	73
`0000000011`	9	73 std+suf	10	28 call:std	1 suf	28 call:std	1 suf	—	—	—	68
`0001`	10	73 non-std	4	24 hash	—	48 call:9	1 suf	—	—	—	77
`011`	11	MULTI-REPORT+73 (RPT73M)	3	16 hash† + 2 × (24 hash + 5 SNR)							77
`00000111`	12	MULTI-73 (73M)	8	16 hash† + 2 × 24 hash							72
`0101`	13	FREE TEXT	4	73 text (Varicode)							77
`00001`	14	*(reserved A)*	5	72 payload							77
`0000010`	15	*(reserved B)*	7	70 payload							77
`00000110`	16	*(reserved C)*	8	69 payload							77

*Type 5 omits suffix fields to pack two standard callsigns, locator, and SNR into 77 bits ($1+28+28+15+5=77$b). †16-bit DX hash (Section IV-D). The "Total" column indicates active information bits; formats with Total <77 bits (Types 1, 2, 8, 9, 12) are right-padded with trailing zeros ($77-$ Total bits) to complete the 77-bit physical frame boundary prior to CRC-14 computation.

Table 4: LQ Varicode assignments for the 104-character free-text alphabet.

Char	Varicode	Len	Freq	Char	Varicode	Len	Freq	Char	Varicode	Len	Freq	Char	Varicode	Len	Freq
␣	`001`	3	16.73%	1	`0000001011`	10	0.09%	*	`000000000110`	13	0.01%	Å	`00000000000010111`	17	<0.01%
E	`010`	3	10.23%	Q	`0000001100`	10	0.07%	[	`000000000111`	14	0.01%	Æ	`00000000000011000`	17	<0.01%
T	`0110`	4	7.44%	!	`0000001101`	10	0.07%	]	`000000000011111`	14	0.01%	Ç	`00000000000011001`	17	<0.01%
A	`0111`	4	6.59%	?	`0000001110`	10	0.07%	|	`000000000000100`	14	0.01%	È	`00000000000011010`	17	<0.01%
O	`1000`	4	6.24%	Z	`0000001111`	10	0.06%	/	`00000000000010100`	15	<0.01%	É	`00000000000011011`	17	<0.01%
N	`1001`	4	5.96%	,	`000000001000`	11	0.06%	$	`000000000000010110`	16	<0.01%	Ê	`00000000000011100`	17	<0.01%
I	`1010`	4	5.90%	.	`000000001001`	11	0.05%	+	`000000000000010111`	16	<0.01%	Ë	`00000000000011101`	17	<0.01%
S	`1011`	4	5.41%	0	`000000001010`	11	0.05%	\n	`000000000000011000`	16	<0.01%	Ì	`00000000000011110`	17	<0.01%
R	`1100`	4	5.00%	2	`000000001011`	11	0.05%	%	`000000000000011001`	16	<0.01%	Í	`00000000000011111`	17	<0.01%
H	`1101`	4	4.76%	8	`000000001100`	11	0.04%	&	`000000000000011010`	16	<0.01%	Î	`00000000000001000`	17	<0.01%
D	`11100`	5	3.48%	3	`000000001101`	11	0.04%	<	`000000000000011011`	16	<0.01%	Ï	`00000000000001001`	17	<0.01%
L	`11101`	5	3.21%	4	`000000001110`	11	0.04%	>	`000000000000111000`	17	<0.01%	Đ	`00000000000001010`	17	<0.01%
C	`11110`	5	2.34%	5	`000000001111`	11	0.04%	@	`000000000000111001`	17	<0.01%	Ñ	`00000000000001011`	17	<0.01%
U	`111110`	6	2.24%	9	`000000000100`	11	0.03%	\	`000000000000111010`	17	<0.01%	Ò	`00000000000001100`	17	<0.01%
M	`111111`	6	2.05%	6	`000000000101`	11	0.03%	^	`000000000000111011`	17	<0.01%	Ó	`00000000000001101`	17	<0.01%
F	`000100`	6	1.95%	7	`000000001100`	12	0.03%	`	`000000000000111100`	17	<0.01%	Ô	`00000000000001110`	17	<0.01%
W	`000101`	6	1.63%	"	`000000001101`	12	0.03%	{	`000000000000111101`	17	<0.01%	Õ	`00000000000001111`	17	<0.01%
P	`000110`	6	1.60%	-	`000000001110`	12	0.03%	}	`000000000000111110`	17	<0.01%	Ö	`00000000000000100`	17	<0.01%
G	`000111`	6	1.56%	'	`000000001111`	12	0.02%	~	`000000000000111111`	17	<0.01%	Ø	`00000000000000101`	17	<0.01%
Y	`000010`	6	1.46%	_	`000000000100`	12	0.02%	¡	`000000000000010000`	17	<0.01%	Ù	`00000000000000110`	17	<0.01%
B	`000011`	6	1.18%	;	`000000000101`	12	0.02%	¿	`000000000000010001`	17	<0.01%	Ú	`00000000000000111`	17	<0.01%
V	`000010`	7	0.85%	:	`000000000110`	12	0.02%	À	`000000000000010010`	17	<0.01%	Û	`00000000000000010`	17	<0.01%
K	`00000110`	8	0.53%	=	`0000000001110`	13	0.02%	Á	`000000000000010011`	17	<0.01%	Ü	`00000000000000011`	17	<0.01%
;	`00000111`	8	0.34%	(	`0000000001111`	13	0.01%	Â	`000000000000010100`	17	<0.01%	Ý	`00000000000000001`	17	<0.01%
X	`000000100`	9	0.16%	)	`0000000000100`	13	0.01%	Ã	`000000000000010101`	17	<0.01%	Þ	`00000000000000000001`	18	<0.01%
J	`0000001010`	10	0.10%	#	`0000000000101`	13	0.01%	Ä	`000000000000010110`	17	<0.01%	[FILL]	`00000000000000000000`	18	<0.01%

IV. FIELD ENCODINGS

To maintain compatibility, LQ adopts core field structures from FT8 [1, 2] (K9AN/K1JT)—28-bit callsigns, 15-bit grid locators, 20-bit CQ modifiers, and 5-bit signal reports—while introducing 1-bit portable suffixes, Base-38 non-standard callsigns, CRC-24/Q hashes, and a 104-character free-text Varicode.

A. Standard Callsign (28 bits)

Adopted from FT8 [1, 2], standard ITU amateur callsigns canonicalize to 6 characters anchored around decimal digit d at position 3: $c_1c_2ds_1s_2s_3$. Prefix (c_1,c_2) right-aligns with space padding in c_1 if 1 character; suffix (s_1,s_2,s_3) left-aligns with spaces. Callsigns matching $c_1c_2ds_1s_2s_3$ without slashes are standard; all others are non-standard (Section IV-C).

Index mappings: $c_1 \in \{␣,0\text{–}9,\text{A–Z}\}$ (radix 37: space $\to 0$, `0-9` $\to 1\ldots10$, `A-Z` $\to 11\ldots36$); $c_2 \in \{0\text{–}9,\text{A–Z}\}$ (radix 36: `0-9` $\to 0\ldots9$, `A-Z` $\to 10\ldots35$); $d \in \{0\text{–}9\}$ (radix 10); and $s_1,s_2,s_3 \in \{␣,\text{A–Z}\}$ (radix 27: space $\to 0$, `A-Z` $\to 1\ldots26$). With indices $(n_1,\ldots,n_6)$, mixed-radix serialization packs into 28-bit integer N_{28}:

$$N_{28} = (n_1\cdot 36\cdot 10 + n_2\cdot 10 + n_3)\cdot 27^3 + n_4\cdot 27^2 + n_5\cdot 27 + n_6. \tag{3}$$

Standard callsigns occupy $0 \le N_{28} \le N_{\max} = 262{,}177{,}559 < 2^{28}$; subsequent states encode tokens `DE` = 262,177,560, `QRZ` = 262,177,561, and `CQ` = 262,177,562. Decoders validate radix ranges to reject corrupted values.

B. Standard Dedicated Suffixes (1 bit)

To support field activations (POTA/SOTA) without hashing, frames reserve an explicit 1-bit suffix: `0`=*none* (e.g., `HB9IPH`), `1`=`/P` (portable).

C. Non-Standard Callsigns (Base-38 Big-Integer Packing)

Non-standard or slashed callsigns pack as Base-38 strings over alphabet ␣ (0), `A–Z` (1–26), `/` (27), and `0–9` (28–37), right-padded with spaces to length L and evaluated big-endian as $N_{\text{b38}} = \sum_{i=0}^{L-1} v_i \cdot 38^{L-1-i}$. **9-Char Callsigns (48 bits):** $38^9 < 2^{48}$ (Types 2, 3, 7, 10), supporting up to 9 characters plus 1-bit portable suffix (`/P`, e.g., `EA6/HB9IP/P`) or slashed calls (`HB9IPH/Q`). **13-Char Callsigns (69 bits):** $38^{13} < 2^{69}$ (Type 4), supporting compound callsigns up to 13 characters plus suffix in CQ (`3B9/HB9IPH/P`). Unnormalized whitespace is rejected.

Alphabet Disambiguation: This Base-38 alphabet (space $\to 0$, `A-Z` $\to 1\ldots26$, `/` $\to 27$, `0-9` $\to 28\ldots37$) places letters before digits, unlike the legacy WSJT-X order (`0-9` $\to 1\ldots10$, `A-Z` $\to 11\ldots36$) used in the 16-bit hash (Section IV-D).

D. Callsign Hashes (24-bit, 20-bit, and 16-bit) & Dehashing

Non-standard stations addressed in directed frames are identified via collision-resistant hashes: **24-Bit Callsign Hash (H_{24}):** Computed over trimmed ASCII callsigns via CRC-24/Q [9] ($G(x)$ = `0x1864CFB`, init/final `0x000000`, non-reflected): $H_{24}(\text{call})$ = CRC-24/Q(call), yielding 2^{24} = 16,777,216 states ($P_{\text{collision}} \le 0.75\%$ across 500 stations). **20-Bit Target Hash (H_{20}):** Used in Type 7 directed replies as 20 MSBs: $H_{20}(\text{call}) = \lfloor H_{24}(\text{call})/2^4 \rfloor = H_{24}(\text{call}) \gg 4$, yielding 2^{20} = 1,048,576 states (e.g., H_{20} = `0x01A4F` for H_{24} = `0x01A4F2`), conserving 4 payload bits for reports while maintaining high collision resistance.

16-Bit DX Callsign Hash (H_{16}): Used in multi-station frames (Types 11 and 12), computed via the WSJT-X 16-bit multiplicative hash [2]. Callsigns right-pad to 11 characters over alphabet (␣ $\to 0$, `0-9` $\to 1\ldots10$, `A-Z` $\to 11\ldots36$, `/` $\to 37$):

$$H_{16}(\text{call}) = \lfloor (\textstyle\sum_{i=0}^{10} v_i \cdot 38^{10-i} \cdot 47055833459)/2^{48} \rfloor \bmod 2^{16}, \tag{4}$$

yielding 65,536 states (e.g., H_{16}(`HB9IPH`) = `0x7A8B`); to prevent collisions, decoders restrict H_{16} dehashing to actively called stations (evicted upon QSO completion or timeout).

UI Resolution: Resolved hashes display bracketed by callsign (e.g., `<HB9IPH>`); unresolved hashes display in hex (e.g., `<01A4F2>`, `<01A4F>`, or `<7A8B>`).

E. Maidenhead Grid Locator (15 bits)

Following the FT8 standard [1, 2, 8], four-character Maidenhead grid locators ($F_1F_2D_1D_2$, with fields $F_1,F_2 \in$ [A–R] and squares $D_1,D_2 \in$ [0–9]) divide the globe into $18\times18 = 324$ fields of $10\times10 = 100$ squares (32,400 total squares). Packed into 15 bits ($N_{\text{grid}} < 32{,}400 < 2^{15} = 32{,}768$):

$$\begin{aligned} N_{\text{grid}} = {} & (F_1 - \text{'A'})\cdot 1800 + (F_2 - \text{'A'})\cdot 100 \\ & + (D_1 - \text{'0'})\cdot 10 + (D_2 - \text{'0'}). \end{aligned} \tag{5}$$

Values $0 \le N_{\text{grid}} \le 32{,}399$ represent valid Maidenhead squares; $N_{\text{grid}} = 32{,}400$ is the blank locator sentinel; values $32{,}401 \le N_{\text{grid}} \le 32{,}767$ are rejected.

F. CQ Modifiers (20 bits)

Directed CQ frames (Types 1 and 3) adopt the 20-bit modifier structure from WSJT-X / FT8 [1, 2] (2^{20} = 1,048,576 states): *1) Unmodified CQ:* $N_{\text{mod}} = 0$ designates general CQ calls. *2) 3-Digit Numeric (000–999):* $1 \le N_{\text{mod}} \le 1000$ conveys numeric designators (e.g., net/channel IDs `CQ 040`), mapped as $N_{\text{mod}} = \text{val} + 1$. *3) Alphanumeric Tokens (≤ 4 chars):* $1001 \le N_{\text{mod}} \le 1{,}048{,}575$ encodes tokens up to 4 characters (e.g., `DX`, `NA`, `POTA`, `SOTA`, `QRP`, `CQWW`; values $> 2^{20} - 1$ rejected). Tokens < 4 characters right-pad with spaces over a 32-symbol alphabet (␣ $\to 0$, `A-Z` $\to 1\ldots26$, `0-4` $\to 27\ldots31$):

$$N_{\text{mod}} = 1000 + \sum_{i=0}^{3} v_i \cdot 32^{3-i}. \tag{6}$$

Omitting digits 5–9 ensures $32^4 = 2^{20}$ fits without overflow; longer modifiers use 4-character abbreviations (`WPX`) or free text.

G. Signal Reports (5 bits)

Following FT8 [1], signal reports convey measured SNR from -26 dB to $+5$ dB via integer $N_{\text{snr}} = \text{clamp}(\text{SNR}_{\text{dB}} + 26, 0, 31) \in [0, 31]$. In directed CALL replies (Types 5–7), reports display without prefix (e.g., `-03`); in acknowledgments (REPORT+73, Types 8 and 11), `R` is prepended (e.g., `R-03`).

H. Free-Text Varicode Encoding (73 bits)

Unstructured text in Type 13 (FREE TEXT, 73 payload bits) uses a 104-character prefix code (Table 4) [7] optimized from character frequencies [10]. The alphabet spans uppercase `A-Z`, digits `0-9`, space, 32 punctuation symbols, 32 Latin-1 characters (¡–Þ), newline `\n`, ↵, and `[FILL]`. Codewords range from 3 to 18 bits (mean 4.25 bits/char), conveying **≈17.2 characters** on average (+32.3% over conventional 13-character payloads). Unused bits are zero-padded; decoders terminate extraction at the first zero bit without emitting fill tokens.

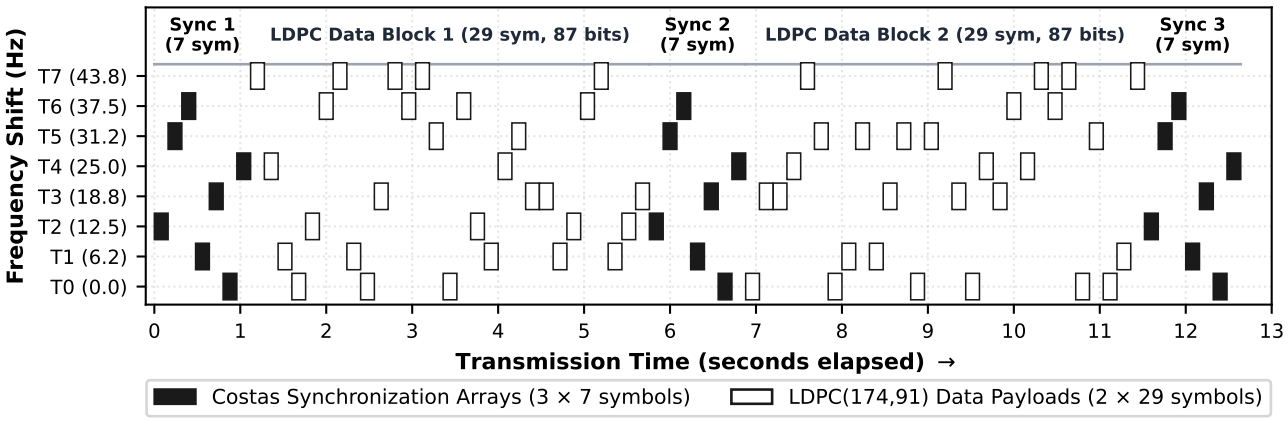


Figure 1: LQ8 physical frame structure (8-GFSK, 50.0 Hz bandwidth, 12.64 s duration).

V. PHYSICAL LAYER

To ensure equal weak-signal sensitivity (−21.0 dB SNR in 2500 Hz bandwidth) and transceiver compatibility, LQ8 deliberately adopts the core physical-layer modulation and error-correction coding of the FT8 protocol [1], engineered by Steve Franke (K9AN) and Joe Taylor (K1JT).

A. Channel Coding: CRC-14 and Systematic LDPC(174,91)

Link-layer encapsulation preserves FT8's channel-coding structure [1]: (1) **77-Bit Payload** from the application layer; (2) **14-Bit CRC Checksum** ($G_{\mathrm{CRC14}}(x)$, Table 5) evaluated by shifting the 77-bit payload MSB-first into a 14-bit LFSR using FT8's generator polynomial `0x2757` followed by 14 trailing zeros, appending remainder $(d_{77}\dots d_{90})$ to form a 91-bit systematic block ($P_{\mathrm{false}} \le 2^{-14} \approx 6.1\times10^{-5}$); 5 zero bits pad to a 12-byte boundary; and (3) **Systematic LDPC(174,91) FEC** utilizing the identical Gallager generator matrix $G_{83\times91}$ (Table 6) [1, 5] producing 83 parity bits ($R \approx 0.523$, 174 codeword bits).

Systematic codeword $\mathbf{c} \in \mathbb{F}_2^{174}$ sets $c_j = d_j$ for $0 \le j < 91$. Parity bits are:

$$c_{91+i} = \bigoplus_{j=0}^{90} G_{i,j}\cdot d_j, \quad 0 \le i < 83, \tag{7}$$

where $G_{83\times91}$ is the Gallager matrix in Table 6 (23 hex digits/row, 91 MSB-aligned bits; bit 0 of digit 23 is zero pad). CRC-14 parameters are listed in Table 5.

Table 5: CRC-14 channel coding parameter specification.

Parameter	Value / Specification
Generator Polynomial $G_{\mathrm{CRC14}}(x)$	$x^{14}+x^{13}+x^{10}+x^9+x^8+x^6+x^4+x^2+x+1$
Hex / LFSR Representation	`0x2757` (14-bit: 10011101010111_2; 15-bit monic: `0x6757`)
Initial State / Final XOR	`0x0000` (14 zeros) / `0x0000` (uninverted)
Input / Output Reflection	`False` / `False` (MSB-first evaluation & output)
Bit Framing & Padding	77 payload bits + 14 CRC bits = 91 systematic bits (+5 pad → 96)
Undetected Error Probability	$P_{\mathrm{false}} \le 2^{-14} \approx 6.10\times10^{-5}$
Invalid Checksum Sentinel	Remainder `0x0000` rejected as corrupt
Target FEC Input Block	Feeds LDPC(174,91) generator $G_{83\times91}$

B. LQ8 Physical Transport & UTC Slot Framing

LQ8 adopts FT8's continuous-phase 8-GFSK transport parameters [1]: 79 symbols in 12.64 s within 15.0 s UTC slots ($R_s = 6.25$ Bd, $\Delta f = 6.25$ Hz, 50.0 Hz footprint, −21.0 dB SNR sensitivity, Figure 1). Transmissions occupy 3.0 kHz USB segments with multiple concurrent subcarrier offsets.

For timestamp t_{UTC}, slot index and parity $s \in \{0,1\}$ evaluate as $k = \lfloor t_{\mathrm{UTC}}/T_{\mathrm{slot}} \rfloor$ and $s = k \bmod 2$ (Even :00/:30 vs. Odd :15/:45). Guard interval $T_{\mathrm{guard}} = T_{\mathrm{slot}} - T_{\mathrm{tx}}$ (2.36 s in LQ8, 2.46 s in LQ4) provides propagation and T/R turnaround margin; stations alternate parity across cycles.

C. Modulation Mapping & Continuous-Phase Synthesis

The 3-bit Gray code maps channel bit triplets to tones: `000`→0, `001`→1, `011`→2, `010`→3, `110`→4, `100`→5, `101`→6, and `111`→7. Continuous-phase audio synthesis $s(t) = A(t)\cos(\phi(t))$ at $f_s = 12{,}000$ Hz employs half-symbol raised-cosine envelope shaping ($T_{\mathrm{ramp}} = T_s/2 = 80.0$ ms in LQ8, 160.0 ms in LQ16) and Gaussian frequency trajectories ($BT = 2.0$):

$$\phi(t) = \phi_0 + 2\pi \int_0^t \left(f_0 + \sum_{k=0}^{N_{\mathrm{sym}}-1} a_k \cdot \Delta f \cdot g(\tau - kT_s) \right) d\tau, \tag{8}$$

where tone index $a_k \in \{0,1,\dots,M-1\}$ is uncentered with tone 0 at base subcarrier f_0 ($\Delta f = 6.25$ Hz in LQ8). Frequency pulse $g(t)$ is truncated to duration $L = 3T_s$ ($t \in [-1.5T_s, +1.5T_s]$):

$$g(t) = \frac{1}{2T_s}\left[\mathrm{erf}\left(2\pi B \frac{t + T_s/2}{\sqrt{2\ln 2}} \right) - \mathrm{erf}\left(2\pi B \frac{t - T_s/2}{\sqrt{2\ln 2}} \right) \right]. \tag{9}$$

D. Costas Synchronization & DSP Pipeline

While LQ8 shares FT8's modulation, CRC-14, and LDPC codes, it deliberately departs in its synchronization structure to achieve physical cross-protocol isolation. Rather than FT8's sync sequence ($[3,1,4,0,6,5,2]$), 8-GFSK modes (LQ8 and LQ16) synchronize via three 7-symbol Costas arrays [6] ($C_7 = [2,5,6,1,3,0,4]$) at symbols 0–6, 36–42, and 72–78. The 174-bit systematic LDPC codeword is partitioned without bit interleaving into two 87-bit data blocks: Data Block 1 (symbols 7–35, $29\times3 = 87$ bits) carries codeword bits $c_0\dots c_{86}$, while Data Block 2 (symbols 43–71) carries $c_{87}\dots c_{173}$. Each symbol maps 3 consecutive bits MSB-first via the 3-bit Gray code. Preambles exhibit ideal autocorrelation with peak sidelobe ≤ 1 and ≤ 1 coincidence hit against FT8, preventing false decodes.

Receiver DSP executes the multi-pass decoding paradigm from WSJT-X [1, 2]: sliding STFT (Hann window, 4-hop oversampling) generating spectral density $S[t_i, f_j]$, 2D matched filter detection across ±0.5 s timing windows, 3-point parabolic frequency refinement ($|\Delta f_{\mathrm{err}}| < 0.1$ Hz), soft LLR extraction, log-domain LDPC belief-propagation decoding (up to 25 passes, terminating early upon $\mathbf{H}\cdot\hat{\mathbf{c}}^T = \mathbf{0}$ and CRC-14 verification), and time-frequency candidate subtraction to decode weaker signals.

Table 6: The complete Gallager LDPC(174,91) generator matrix $G_{83\times91}$ (83 rows × 23 hex digits). Each row represents 92 bits MSB-aligned; bit 0 of the 23rd hex digit is an unused zero padding bit.

Row	Generator Vector (23 Hex Digits)	Row	Generator Vector (23 Hex Digits)	Row	Generator Vector (23 Hex Digits)
00	8329ce11bf31eaf509f27fc	28	8eba1a13db3390bd6718cec	56	7ee18230c583cccc57d4b08
01	761c264e25c259335493132	29	753844673a27782cc42012e	57	a066cb2fedafc9f52664126
02	dc265902fb277c6410a1bdc	30	06ff83a145c37035a5c1268	58	bb23725abc47cc5f4cc4cd2
03	1b3f417858cd2dd33ec7f62	31	3b37417858cc2dd33ec3f62	59	ded9dba3bee40c59b5609b4
04	09fda4fee04195fd034783a	32	9a4a5a28ee17ca9c324842c	60	d9a7016ac653e6decdc9036
05	077cccc11b8873ed5c3d48a	33	bc29f465309c977e89610a4	61	9ad46aed5f707f280ab5fc4
06	29b62afe3ca036f4fe1a9da	34	2663ae6ddf8b5ce2bb29488	62	e5921c77822587316d7d3c2
07	6054faf5f35d96d3b0c8c3e	35	46f231efe457034c1814418	63	4f14da8242a8b86dca73352
08	e20798e4310eed27884ae90	36	3fb2ce85abe9b0c72e06fbe	64	8b8b507ad467d4441df770e
09	775c9c08e80e26ddae56318	37	de87481f282c153971a0a2e	65	22831c9cf1169467ad04b68
10	b0b811028c2bf997213487c	38	fcd7ccf23c69fa99bba1412	66	213b838fe2ae54c38ee7180
11	18a0c9231fc60adf5c5ea32	39	f0261447e9490ca8e474cec	67	5d926b6dd71f085181a4e12
12	76471e8302a0721e01b12b8	40	4410115818196f95cdd7012	68	66ab79d4b29ee6e69509e56
13	ffbccb80ca8341fafb47b2e	41	088fc31df4bfbde2a4eafb4	69	958148682d748a38dd68baa
14	66a72a158f9325a2bf67170	42	b8fef1b6307729fb0a078c0	70	b8ce020cf069c32a723ab14
15	c4243689fe85b1c51363a18	43	5afea7acccb77bbc9d99a90	71	f4331d6d461607e95752746
16	0dff739414d1a1b34b1c270	44	49a7016ac653f65ecdc9076	72	6da23ba424b9596133cf9c8
17	15b48830636c8b99894972e	45	1944d085be4e7da8d6cc7d0	73	a636bcbc7b30c5fbeae67fe
18	29a89c0d3de81d665489b0e	46	251f62adc4032f0ee714002	74	5cb0d86a07df654a9089a20
19	4f126f37fa51cbe61bd6b94	47	56471f8702a0721e00b12b8	75	f11f106848780fc9ecdd80a
20	99c47239d0d97d3c84e0940	48	2b8e4923f2dd51e2d537fa0	76	1fbb5364fb8d2c9d730d5ba
21	1919b75119765621bb4f1e8	49	6b550a40a66f4755de95c26	77	fcb86bc70a50c9d02a5d034
22	09db12d731faee0b86df6b8	50	a18ad28d4e27fe92a4f6c84	78	a534433029eac15f322e34c
23	488fc33df43fbdeea4eafb4	51	10c2e586388cb82a3d80758	79	c989d9c7c3d3b8c55d75130
24	827423ee40b675f756eb5fe	52	ef34a41817ee02133db2eb0	80	7bb38b2f0186d46643ae962
25	abe197c484cb74757144a9a	53	7e9c0c54325a9c15836e000	81	2644ebadeb44b9467d1f42c
26	2b500e4bc0ec5a6d2bdbdd0	54	3693e572d1fde4cdf079e86	82	608cc857594bfbb55d69600
27	c474aa53d70218761669360	55	bfb2cec5abe1b0c72e07fbe		

VI. EXTENDED MODES: LQ16, LQ4, AND LQ2

While **LQ8** is the primary mode for 15-second scheduled weak-signal communications, the 77-bit variable-length architecture scales across three complementary physical profiles (Table 7):

- **LQ16 (Slow / Ultra Weak):** 8-GFSK mode for severe weak-signal conditions (−24.0 dB SNR threshold in 2500 Hz) with 320 ms symbol period (25.0 Hz bandwidth, $BT = 2.0$, 25.28 s duration) for extreme weak-signal and EME paths.
- **LQ4 (Fast / Contest):** 4-GFSK mode for rapid contesting and pileups (30.0 s contact, 120 QSOs/h standard, 480 QSOs/h Fox/Hound) with 48 ms symbol period, 83.3 Hz bandwidth, and 5.04 s transmission in 7.5 s UTC slots.
- **LQ2 (Turbo Burst):** 4-GFSK high-throughput profile for dynamic burst channels (15.0 s full contact, 240 QSOs/h standard, 960 QSOs/h Fox/Hound) with 24 ms symbol period, 166.7 Hz bandwidth, and 2.52 s transmission in 3.75 s slots for meteor scatter.

A. 4-GFSK Frame Structure & Modulation

LQ4 and LQ2 transmit 105 symbols/frame using continuous-phase 4-GFSK ($BT = 1.0$, tone spacing $\Delta f = 1/T_s$, modulation index $h = 1.0$):

- **Consolidated Framing Sequence:** The 105-symbol frame is partitioned into: tone-0 ramp-up (sym 0, $T_{\text{ramp}} = T_s/2$, 24 ms in LQ4, 12 ms in LQ2); Costas sync $S_1 = [0,2,3,1]$ (syms 1–4); Data Block 1 (syms 5–33, 29 symbols, codeword bits 0…57); Costas $S_2 = [1,3,2,0]$ (syms 34–37); Data Block 2 (syms 38–66, 29 symbols, bits 58…115); Costas $S_3 = [2,0,1,3]$ (syms 67–70); Data Block 3 (syms 71–99, 29 symbols, bits 116…173); Costas $S_4 = [3,1,0,2]$ (syms 100–103); and tone-0 ramp-down (sym 104, $T_{\text{ramp}} = T_s/2$).
- **Scrambling & Whitening:** Prior to CRC-14/LDPC channel encoding, the 77-bit payload is bitwise XORed with the 10-byte whitening sequence (`4A 5E 89 B4 B0 8A 79 55 BE 28`$_{16}$) to suppress discrete spectral lines. Bits 77–79 (byte 9, `0x28`) are zeros prior to CRC-14 calculation.
- **Data Framing & Gray Coding:** Codeword bits are mapped sequentially without bit interleaving. The three 29-symbol data blocks map consecutive pairs (b_{2k}, b_{2k+1}) of the 174-bit codeword to 4-GFSK tones via Gray coding (`00`→0, `01`→1, `11`→2, `10`→3).
- **Ramping & Guard Margins:** Tone-0 symbols at indices 0 and 104 frame the sequence (5.04 s for LQ4, 2.52 s for LQ2). Guard intervals (2.46 s in LQ4, 1.23 s in LQ2) absorb propagation delay and T/R turnaround.

B. Profile Selection and Channel Matching

Transceivers dynamically select operational profiles according to ionospheric channel coherence time T_c and link signal-to-noise ratio:

- When SNR < -21.0 dB and Doppler spread $\sigma_f \leq 0.5$ Hz, LQ16 provides a +3.0 dB gain over LQ8, sustaining readability down to −24.0 dB for extreme paths.
- In rapid-fading channels ($T_c < 10$ s), LQ4's 48 ms symbols and 7.5 s slots ensure robust sync, completing verified contacts in 30.0 s before phase degrades.
- For dynamic burst channels (e.g., meteor scatter), LQ2's 2.52 s bursts capture transient trails, completing 4-slot exchanges in 15.0 s.

Table 7: Multi-mode parameter and performance comparison across standard reference modes and the LQ family.

Parameter	FT8 (Ref)	FT4 (Ref)	LQ8	LQ16	LQ4	LQ2
Modulation	8-GFSK	4-GFSK	8-GFSK	8-GFSK	4-GFSK	4-GFSK
Bandwidth	50.0 Hz	83.3 Hz	50.0 Hz	25.0 Hz	83.3 Hz	166.7 Hz
Time slot	15.0 s	7.5 s	15.0 s	30.0 s	7.5 s	3.75 s
Transmission time	12.64 s	5.04 s	12.64 s	25.28 s	5.04 s	2.52 s
Sensitivity (SNR)	−21.0 dB	−17.5 dB	−21.0 dB	−24.0 dB	−17.5 dB	−14.0 dB
Relative RF power req.	**0 dB (ref)**	+3.5 dB (2.2×)	0 dB (ident.)	−3.0 dB (0.5×)	+3.5 dB (2.2×)	+7.0 dB (5.0×)
Type coding	Fixed	Fixed	Huffman	Huffman	Huffman	Huffman
QSO messages	5–6	5–6	4	4	4	4
QSO duration	75.0–90.0 s	37.5–45.0 s	60.0 s	120.0 s	30.0 s	15.0 s
Standard QSO Rate*	40–48/h	80–96/h	60/h	30/h	120/h	240/h
Pileup Rate (1-carrier)*	60/h	120/h	240/h	120/h	480/h	960/h
Pileup Rate (2-carrier)*	120/h	240/h	480/h	240/h	960/h	1920/h
Standard Suffixes	/P, /R	/P, /R	/P	/P	/P	/P
Max non-std callsign	11 chars	11 chars	13 + std suffix	13 + std suffix	13 + std suffix	13 + std suffix
Callsign hash length	10/12/22-b	10/12/22-b	24/20/16-b	24/20/16-b	24/20/16-b	24/20/16-b
Free-text alphabet	42 chars	42 chars	104 chars	104 chars	104 chars	104 chars
Free-text capacity	71 bits (∼13 c)	71 bits (∼13 c)	73 bits (∼17 c)	73 bits (∼17 c)	73 bits (∼17 c)	73 bits (∼17 c)

*FT8/FT4 contacts require 5–6 messages (75–90 s / 37.5–45 s) using `RR73` or `RRR`/73; LQ completes contacts in 4 messages (60 s in LQ8). Peak rates assume continuous zero-loss operation. Pipelined pileup rates reflect F/H mode confirming 1 station/cycle in FT8/FT4 vs. 2 in LQ (Type 11).

VII. CONCLUSION

LQ8 demonstrates that continuous-phase 8-GFSK and systematic LDPC$(174, 91)$ transport deliver higher throughput through variable-length prefix framing while maintaining full backward physical compatibility and −21.0 dB SNR sensitivity in a 2500 Hz reference bandwidth. By optimizing message prefix allocations to match operational entropy, LQ8 packs callsigns, locators, and reports into a 77-bit budget, finalizing confirmed two-way contacts in 4 transmission slots (60 s). For high-density pileups, LQ8 introduces the single-carrier MULTI-REPORT+73 protocol, confirming two answering stations simultaneously without RF power-splitting penalties and achieving throughputs up to 240 QSOs/h (480 QSOs/h in dual-carrier operation).

The unified protocol family extends this architecture across three complementary physical profiles. LQ16 scales symbol duration to 320 ms, delivering a +3.0 dB sensitivity advantage (−24.0 dB SNR threshold) across ultra-weak fading and EME paths. For rapid contesting and dynamic propagation, LQ4 (7.5 s slots) and LQ2 (3.75 s slots) utilize 4-GFSK modulation and Costas sync arrays to complete contacts in 30.0 s and 15.0 s, supporting burst rates up to 960 QSOs/h on VHF/UHF meteor scatter and sporadic-E openings.

The complete software suite is implemented in the open-source C++ reference library (`lq_lib`) under the MIT License, featuring an automated verification harness achieving $> 90\%$ line coverage alongside comprehensive golden vector test suites across all operational profiles. End-to-end field verification has been demonstrated in real-world portable activations via **qFT8** for Android across HF bands. Bit-exact codec traces for all 13 message formats are verified under AWGN, Rayleigh fading, and clock drift; reference sensitivities span −14.0 dB (LQ2) to −24.0 dB (LQ16).

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